

Business Requirement Document

Customer Reviews Module

ShopSphere E-Commerce Platform — Platform Modernization Program (FY2026)

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1. Document Information

Program	ShopSphere Platform Modernization Program (FY2026)
Platform	ShopSphere E-Commerce Platform (Web + Mobile App), ~2 million MAU across India, UAE, and Southeast Asia
Module	Customer Reviews & Ratings
Prepared For	Rajesh Mehta, VP of E-Commerce (Executive Sponsor)
Distribution	Product, Business Analysis, IT/Architecture, QA, Design, Risk & Compliance

2. Executive Summary

ShopSphere Technologies Pvt. Ltd. operates the ShopSphere E-Commerce Platform, a B2C marketplace serving approximately 2 million monthly active users across India, UAE, and Southeast Asia. Customer-generated reviews and ratings are among the strongest influences on purchase conversion on the Product Detail Page (PDP), yet the current post-purchase feedback experience is fragmented: review prompts are inconsistent, moderation is largely manual, "Verified Purchase" labeling is unreliable, and there is no structured mechanism for sellers to respond to customer feedback or for shoppers to ask pre-purchase questions. This document defines the business and functional requirements for a modernized Customer Reviews Module under the ShopSphere Platform Modernization Program (FY2026). The module will deliver automated post-delivery review prompts, a rich review submission experience (star rating, text, photo/video), an automated-plus-manual content moderation workflow powered by the SafeContent AI Moderation API, authenticated "Verified Purchase" badging, community helpful/unhelpful voting, a product Q&A section, and seller/brand response capability. The module is designed to increase review volume and authenticity while protecting the platform from fraudulent, incentivized, or abusive content, and to feed sentiment signals into Salesforce CRM for proactive customer service escalation.

3. Business Objective

The primary business objective is to increase trust-driven purchase conversion and reduce return rates by making product reviews more abundant, more authentic, and more actionable. Specific objectives include: (a) increasing the volume of post-delivery reviews collected through automated, well-timed prompts; (b) ensuring only genuine, purchase-verified feedback is prominently badged to combat review fraud; (c) reducing the operational burden of manual moderation through AI-assisted screening while maintaining a strict moderation SLA; (d) surfacing aggregate ratings and review counts on the PDP and PLP to improve shopper decision-making and reduce cart abandonment; (e) enabling sellers and brand teams to respond publicly to reviews, improving perceived responsiveness; and (f) routing highly negative review sentiment into Salesforce CRM so that Customer Service can proactively intervene before dissatisfaction escalates into returns, refunds, or public complaints.

4. Project Scope

In Scope

- Automated post-delivery review and rating prompts delivered via the Notification Service (email/SMS/push), typically 5-7 days after confirmed delivery.
- Review submission experience: star rating (1-5), free-text review, and photo/video attachment upload.
- Automated content moderation via the SafeContent AI Moderation API, supplemented by a manual moderation queue for escalations.
- "Verified Purchase" badge logic tied to confirmed order and delivery records.
- Helpful/unhelpful community voting on published reviews.
- Product Q&A section allowing shoppers to ask questions and receive answers from other customers, sellers, or brand representatives.
- Seller/brand response capability to publicly reply to customer reviews.
- Aggregate rating computation and display on the PDP and PLP.

Out of Scope

- Influencer and paid review campaign management.
- Third-party review syndication (e.g., importing reviews from other marketplaces).
- Review-linked loyalty rewards program.

5. Stakeholders

Name	Role	Department	Responsibility
Rajesh Mehta	VP of E-Commerce	Business	Executive Sponsor
Ananya Sharma	Product Owner, ShopSphere Platform	Product	Prioritization & sign-off
Lucas Bennett	Business Analyst, Platform Operations	Business Analysis	Requirements owner
Sarah Chen	Enterprise IT Architect	IT/Architecture	Technical feasibility
Michael Torres	QA Lead	Quality Assurance	Test strategy & sign-off
Priya Nair	UX Lead	Design	User experience
Fatima Al-Sayed	Compliance & Security Officer	Risk & Compliance	Content moderation policy

6. Business Requirements

ID	Business Requirement
BR-001	The platform shall solicit product reviews from customers automatically after confirmed delivery to increase the volume and freshness of review content available to future shoppers.
BR-002	The platform shall visibly distinguish reviews from customers with a confirmed, verified purchase to increase shopper trust and reduce the influence of fraudulent or incentivized reviews.
BR-003	The platform shall screen all submitted review content (text, photo, video) for profanity, spam, and policy violations prior to public display in order to maintain a safe and compliant marketplace.
BR-004	The platform shall allow the shopping community to rate the helpfulness of published reviews so that the most useful feedback surfaces to future buyers.
BR-005	The platform shall enable sellers and brand representatives to publicly respond to customer reviews to demonstrate responsiveness and resolve concerns transparently.
BR-006	The platform shall provide a product Q&A capability so prospective buyers can obtain pre-purchase clarifications from other customers or sellers, reducing pre-sale support inquiries.
BR-007	The platform shall route reviews carrying strongly negative sentiment into Salesforce CRM to trigger proactive customer service outreach and reduce escalation to returns or public complaints.
BR-008	The platform shall aggregate individual ratings into a single product-level rating displayed on the PDP and PLP to support faster, more confident purchase decisions.

7. Functional Requirements

Review Submission

- **FR-001** The system shall trigger a review-request notification (email, SMS, or push, per customer preference) 5-7 days after the order status changes to "Delivered."
- **FR-002** The system shall allow a customer to submit a review consisting of a mandatory 1-5 star rating, an optional free-text comment (up to 2,000 characters), and up to 6 photo or 1 video attachment(s) per review.
- **FR-003** The system shall restrict review submission to one review per customer per delivered order line item (SKU), with the ability to edit the review within 30 days of original submission.
- **FR-004** The system shall capture the order ID and delivery confirmation timestamp at the time of review submission to support Verified Purchase eligibility checks.

Moderation

- **FR-005** The system shall submit all review text and media to the SafeContent AI Moderation API immediately upon submission for automated screening against profanity, spam, hate speech, and PII exposure policies.
- **FR-006** The system shall automatically publish reviews that pass automated moderation with a confidence score at or above the configured threshold (default 90%), and shall route reviews below the threshold to a manual moderation queue.
- **FR-007** The system shall allow authorized moderators to approve, reject, or edit-and-approve (for redaction of PII only) queued reviews, recording the moderator ID and decision timestamp.
- **FR-008** The system shall allow any logged-in user to report/flag a published review as inappropriate, automatically re-queuing it for manual re-review after a configurable number of flags (default 3) or a single flag from a Trust & Safety-tagged account.

Display & Aggregation

- **FR-009** The system shall compute and display an aggregate star rating (weighted average to one decimal place) and total review count on the PDP and PLP, refreshed within 15 minutes of new review publication.
- **FR-010** The system shall display a "Verified Purchase" badge on any published review linked to a confirmed and delivered order for the exact SKU being reviewed.
- **FR-011** The system shall allow logged-in users to mark a published review as "Helpful" or "Unhelpful," display the net helpful count, and allow sorting of reviews by Most Helpful, Most Recent, and Highest/Lowest Rating.
- **FR-012** The system shall provide a rating distribution breakdown (percentage of 5-star, 4-star, 3-star, 2-star, and 1-star reviews) on the PDP.

Q&A and Seller Response

- **FR-013** The system shall allow any logged-in user to post a product-related question on the PDP, visible to other shoppers, sellers, and brand representatives.
- **FR-014** The system shall allow other customers, the fulfilling seller, or verified brand representatives to submit answers to posted questions, subject to the same content moderation workflow as reviews.
- **FR-015** The system shall allow a seller or brand representative to submit one public response per customer review, displayed immediately below the original review.
- **FR-016** The system shall push a summary of newly submitted reviews with a rating of 1 or 2 stars to Salesforce CRM within 5 minutes of publication, tagged against the corresponding customer profile for service follow-up.

8. Non-Functional Requirements

Performance

- **NFR-001** Review and Q&A submission APIs shall respond within 2 seconds at the 95th percentile under normal load.
- **NFR-002** PDP review and rating widgets shall load within 2 seconds at the 95th percentile, including aggregate rating and first page of reviews.

Security

- **NFR-003** All review-related APIs and services shall comply with the OWASP Top 10 secure coding guidelines.

- **NFR-004** Review content, attachments, and reviewer PII shall be encrypted using AES-256 at rest and TLS 1.2 or higher in transit.
- **NFR-005** All review and Q&A submissions shall pass through content moderation to prevent abusive, fraudulent, or policy-violating content prior to public exposure.

Availability

- **NFR-006** The Customer Reviews Module shall maintain 99.9% uptime for all customer-facing submission, display, and voting functions.

Scalability

- **NFR-007** The module shall support up to 8x normal review submission volume during peak sale events (e.g., Diwali, Black Friday) without degradation of response times beyond defined thresholds.
- **NFR-008** The moderation pipeline shall auto-scale to process surge volumes without increasing the automated moderation SLA beyond 2 minutes per submission.

Reliability

- **NFR-009** Review and Q&A data shall be protected with a Recovery Point Objective (RPO) of 15 minutes and a Recovery Time Objective (RTO) of 1 hour.
- **NFR-010** No submitted review, attachment, or vote shall be lost in the event of a moderation service outage; submissions shall queue durably and resume processing on service recovery.

9. User Stories

- **US-001** As a customer who received my order, I want to receive a review prompt a few days after delivery, so that I can easily share my experience while it is still fresh.
- **US-002** As a customer, I want to attach photos or a video to my review, so that other shoppers can see the real product condition and usage.
- **US-003** As a shopper browsing a product, I want to see a "Verified Purchase" badge on reviews, so that I can trust the feedback is from a genuine buyer.
- **US-004** As a shopper, I want to mark reviews as helpful or unhelpful, so that the most useful feedback rises to the top for other buyers.
- **US-005** As a prospective buyer, I want to ask a question about a product before purchasing, so that I can clarify details not covered in the product description.
- **US-006** As a seller, I want to respond publicly to a customer review, so that I can address concerns and show responsiveness to the community.
- **US-007** As a Compliance & Security Officer, I want all review content automatically screened for abusive or fraudulent content, so that the platform remains safe and policy-compliant.
- **US-008** As a Customer Service agent, I want to be alerted in Salesforce CRM when a customer posts a very negative review, so that I can proactively reach out and resolve the issue.

10. Acceptance Criteria

AC ID	Related ID	Criteria	Expected Result
AC-001	FR-001	Given an order marked "Delivered," when 5-7 days elapse, then a review-request notification is sent.	Customer receives email/SMS/push prompting a review.
AC-002	FR-002	Given a customer opens the review form, when they submit a rating with text and photos, then the review is saved and queued for moderation.	Review record created with status "Pending Moderation."
AC-003	FR-006	Given a review is submitted, when SafeContent AI returns a confidence score $\geq 90\%$, then the review is auto-published.	Review appears on PDP within 2 minutes without manual intervention.
AC-004	FR-006	Given a review scores below 90% confidence, when moderation completes, then the review routes to the manual queue.	Review status set to "In Manual Review," not publicly visible.
AC-005	FR-010	Given a review is linked to a delivered order for the reviewed SKU, when the review publishes, then a Verified Purchase badge is attached.	Badge visibly displayed next to reviewer name.
AC-006	FR-008	Given a published review receives 3 user flags, when the threshold is reached, then the review is automatically re-queued for manual review.	Review temporarily hidden pending moderator decision.

AC-007	US-004	Given a shopper clicks "Helpful" on a review, when the vote is recorded, then the helpful count updates in real time.	Updated count visible without page reload.
AC-008	FR-016	Given a review is published with a 1 or 2 star rating, when publication completes, then a record is pushed to Salesforce CRM within 5 minutes.	CRM case or flag created against customer profile.
AC-009	FR-015	Given a seller opens a review on their product, when they submit a response, then the response displays publicly beneath the review.	Seller response visible to all shoppers viewing the review.
AC-010	FR-009	Given a new review is published, when the aggregation job runs, then the PDP aggregate rating and count update within 15 minutes.	PDP reflects updated average rating and review count.

11. Business Rules

- **RULE-001** A "Verified Purchase" badge shall only be granted when the reviewer's account has a confirmed, delivered order for the exact SKU being reviewed, excluding cancelled or fully refunded orders.
- **RULE-002** A customer may submit only one review per delivered order line item; subsequent submission attempts for the same line item shall be treated as an edit to the existing review.
- **RULE-003** Reviews and Q&A content must pass SafeContent AI Moderation API screening before becoming publicly visible; no review shall bypass moderation, including from Verified Purchase reviewers.
- **RULE-004** Reviews flagged as spam, incentivized (e.g., referencing a discount or refund in exchange for a positive rating), or otherwise fraudulent shall be rejected and shall not count toward the product's aggregate rating.
- **RULE-005** Manual moderation decisions on flagged or queued content must be completed within the moderation SLA defined in Section 20; unresolved items beyond SLA shall auto-escalate to the Compliance & Security Officer.
- **RULE-006** Sellers and brand representatives may submit only one official response per review and may not edit or delete a customer's original review content.
- **RULE-007** Reviews rejected during moderation shall not be visible to any user other than the original submitter, who shall receive an in-app notification with the rejection reason.
- **RULE-008** The aggregate product rating shall exclude reviews in "Pending Moderation," "Rejected," or "Removed" status.

12. Process Flow Description

Primary Flow — Review Submission to Publish:

1. Order status changes to "Delivered" in the Order Management system.
2. 5-7 days post-delivery, the Notification Service sends a review-request prompt via the customer's preferred channel (email, SMS, or push).
3. The customer opens the review form from the notification link or from their Order History page and submits a star rating, optional text, and optional photo/video attachments.
4. The system validates order eligibility and attaches Verified Purchase status where applicable.
5. The review is submitted to the SafeContent AI Moderation API for automated screening of text and media.
6. If the confidence score meets or exceeds the auto-publish threshold, the review is published immediately and the aggregate rating recalculation job is triggered.
7. If the score falls below the threshold, the review is routed to the manual moderation queue for human review by an authorized moderator.
8. The moderator approves, rejects, or approves-with-redaction the review; the decision and moderator ID are logged.
9. Upon publication, the review becomes visible on the PDP, contributes to the aggregate rating, and becomes eligible for helpful/unhelpful voting and seller response.
10. If the published rating is 1 or 2 stars, a sentiment record is pushed to Salesforce CRM to trigger proactive customer service follow-up.

Alternate/Exception Flow — Review Flagged During Moderation:

1. A published review receives community flags or a single flag from a Trust & Safety-tagged account.
2. Once the flag threshold is reached, the system automatically hides the review from public view and re-queues it for manual review.
3. A moderator reviews the flagged content against platform policy and either reinstates, edits (PII redaction only), or permanently removes the review.

4. The original submitter is notified of the outcome; if removed, the review is excluded from the aggregate rating recalculation.
5. Repeated policy violations by the same reviewer account are logged and escalated to the Compliance & Security Officer for potential account-level action.

13. Assumptions

- All customers submitting reviews have an authenticated ShopSphere account linked to their order history.
- The SafeContent AI Moderation API is available as a licensed third-party service with contracted uptime and response-time commitments.
- The Notification Service already supports templated, multi-channel (email/SMS/push) delivery and customer channel preferences.
- Salesforce CRM is the system of record for customer profiles and service case management across ShopSphere markets.
- Order and delivery confirmation data is reliably available from the Order Management and Logistics systems in near real time.
- Sellers and brand representatives have existing authenticated portal access through which review responses can be submitted.

14. Constraints

- The module must operate within existing PDP page-load performance budgets and must not increase PDP load time beyond the 2-second (95th percentile) NFR target.
- Content moderation decisions must comply with local content and consumer protection regulations across India, UAE, and Southeast Asia, which may differ in permissible content standards.
- Photo/video attachment storage is capped at a defined per-review size limit to control cloud storage costs.
- The manual moderation queue is staffed by a limited Trust & Safety team, constraining the volume of items that can be reviewed within SLA during peak events without additional automation or staffing.

15. Dependencies

- Order Management and Logistics systems for accurate delivery confirmation timestamps used in Verified Purchase eligibility and review-prompt timing.
- Notification Service for multi-channel delivery of review-request prompts.
- SafeContent AI Moderation API for automated text/image/video content screening.
- Salesforce CRM for linking review sentiment to customer profiles and triggering service case escalation.
- Product Catalog service for surfacing aggregate ratings and review counts on the PDP and PLP.

16. Risks

Risk ID	Description	Likelihood	Impact	Mitigation
RISK-001	Fraudulent or incentivized reviews undermine platform trust and rating accuracy.	Medium	High	Enforce Verified Purchase gating, AI fraud pattern detection, and manual audit sampling.
RISK-002	Moderation backlog during peak sale events causes SLA breaches and delayed publication.	High	Medium	Auto-scale moderation pipeline and pre-stage additional Trust & Safety staffing for peak events.
RISK-003	SafeContent AI Moderation API outage halts automated screening.	Low	High	Durable submission queueing with automatic reprocessing on service recovery; fallback manual review.
RISK-004	Low review submission rates limit the volume of trust-building content on the PDP.	Medium	Medium	Optimize notification timing/channel and simplify the submission form based on UX testing.
RISK-005	Negative reviews are not escalated to Customer Service in time, leading to unresolved dissatisfaction.	Medium	High	Automate the CRM push for 1-2 star reviews within 5 minutes of publication with monitoring alerts.
RISK-006	Inconsistent content policy enforcement across India, UAE, and Southeast Asia leads to regulatory or	Low	High	Establish region-specific moderation policy overlays reviewed by the Compliance &

17. Data Requirements

The module captures and stores review, rating, Q&A, and moderation data with defined retention aligned to platform data governance policy. Review and Q&A records, including associated PII, are retained for the lifetime of the associated customer account plus 3 years, or per applicable local regulatory requirements, whichever is longer, after which data is anonymized or purged.

Data Field	Description
Review ID	Unique identifier for each submitted review.
Customer ID	Identifier of the submitting customer, linked to their ShopSphere account.
Order ID / SKU	Order line item and SKU associated with the review, used for Verified Purchase eligibility.
Rating	Star rating value from 1 to 5.
Review Text	Free-text comment, up to 2,000 characters.
Attachments	Photo/video file references (up to 6 photos or 1 video per review).
Verified Purchase Flag	Boolean indicating confirmed, delivered purchase of the reviewed SKU.
Moderation Status	Pending / Approved / Rejected / Flagged / Removed.
Helpful/Unhelpful Counts	Aggregate vote counts from the shopper community.
Seller Response	Text of the seller/brand reply, if submitted.
Timestamps	Submission, moderation decision, and last-edited timestamps.

18. Integration Requirements

System	Purpose	Direction	Data Exchanged
Salesforce CRM	Link review sentiment to customer profile and trigger service case escalation for very negative reviews.	Outbound (Reviews → CRM)	Customer ID, review rating, review text summary, sentiment flag.
Notification Service	Deliver post-delivery review-request prompts and moderation outcome notifications.	Outbound (Reviews → Notification Service)	Customer ID, order ID, notification channel preference, template data.
SafeContent AI Moderation API	Automated screening of review/Q&A text and media for profanity, spam, and policy violations.	Bidirectional	Submitted content payload out; moderation confidence score and violation category in.
Product Catalog	Surface aggregate rating and review count on the PDP and PLP.	Outbound (Reviews → Catalog)	SKU, aggregate rating, review count, rating distribution.
Analytics/BI (Tableau, GA4)	Reporting and dashboarding on review volume, sentiment, and moderation performance.	Outbound (Reviews → Analytics)	Review events, moderation outcomes, engagement metrics.

19. Reporting Requirements

- Daily dashboard of review submission volume by category, region, and channel.
- Moderation performance report showing automated vs. manual approval rates and average turnaround time against SLA.
- Verified Purchase ratio report tracking the proportion of badged versus unbadged reviews per category.
- Flagged/removed content report for the Compliance & Security Officer, including repeat-offender account tracking.
- Product-level rating trend report highlighting SKUs with declining average ratings for merchandising follow-up.
- Q&A responsiveness report tracking time-to-answer for customer and seller responses.
- Negative review-to-CRM escalation report tracking case creation and resolution outcomes.

20. Success Metrics

Metric	Target
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Post-delivery review submission rate	≥ 18% of delivered orders within 30 days of prompt (up from current baseline of ~7%)
Verified Purchase share of published reviews	≥ 85%
Automated moderation turnaround (auto-publish path)	≤ 2 minutes per submission
Manual moderation SLA compliance	≥ 95% of queued items resolved within 24 hours
Reduction in confirmed fraudulent/incentivized reviews	≥ 40% reduction year-over-year
CRM escalation response time for 1-2 star reviews	Initial customer service contact within 24 hours
PDP conversion uplift attributable to review presence	≥ 5% increase for SKUs with 10+ reviews vs. SKUs with none

21. Future Enhancements

- Introduce a review-linked loyalty rewards program to incentivize high-quality, non-fraudulent feedback (currently out of scope).
- Support third-party review syndication to import ratings from partner marketplaces.
- Add AI-generated review summaries highlighting common themes across large volumes of reviews for a product.
- Enable video-based Q&A responses from sellers for complex product queries.
- Introduce sentiment-based dynamic sorting that surfaces the most relevant reviews per shopper's stated concerns (e.g., "durability," "sizing").

22. Glossary

Term	Definition
SKU	Stock Keeping Unit — a unique identifier for a distinct sellable product variant.
PDP	Product Detail Page — the page displaying full product information, pricing, and reviews.
PLP	Product Listing Page — the page displaying a list/grid of products, including aggregate ratings.
Verified Purchase	A badge indicating the reviewer has a confirmed, delivered order for the reviewed SKU.
Moderation SLA	The maximum allowable time for automated or manual moderation to reach a publish/reject decision.
PII	Personally Identifiable Information — data that can identify a specific individual.
GMV	Gross Merchandise Value — total value of merchandise sold through the platform over a period.
SLA	Service Level Agreement — a defined commitment on service performance or turnaround time.
Q&A	Question and Answer — the product page feature allowing shoppers to ask and receive answers to product questions.
RMA	Return Merchandise Authorization — the process governing product returns.
B2C	Business-to-Consumer — a commerce model where a business sells directly to individual consumers.

23. Appendix

Reference Documents: ShopSphere Content Moderation Policy v2.1; ShopSphere Notification Service Integration Guide; SafeContent AI Moderation API Specification v3.0; ShopSphere Data Retention & Privacy Policy.

Related BRDs: BRD-EC-ORD-003 Order Management Module; BRD-EC-NTF-005 Notification Service Module; BRD-EC-CAT-002 Product Catalog Module; BRD-EC-CRM-011 Customer Service & CRM Integration Module.

Version	Date	Author	Change Description
0.1	10-Feb-2026	Lucas Bennett	Initial draft circulated for stakeholder review.
0.9	25-Feb-2026	Lucas Bennett	Incorporated feedback from Product, UX, and Compliance stakeholders.
1.0	05-Mar-2026	Lucas Bennett	Finalized and approved by Rajesh Mehta and Ananya Sharma.